Online Book Store Data Analysis Using SQL

Basic queries:

1) Retrieve all books in the "Fiction" genre:

```
SELECT *
FROM books
WHERE Genre = "Fiction";
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

Book_ID	Title	Author	Genre	Published_Year	Price	Stock
4	Customizable 24hour product	Christopher Andrews	Fiction	2020	43.52	8
22	Multi-layered optimizing migration	Wesley Escobar	Fiction	1908	39.23	78
28	Expanded analyzing portal	Lisa Coffey	Fiction	1941	37.51	79
29	Quality-focused multi-tasking challenge	Katrina Underwood	Fiction	1905	31.12	100
31	Implemented encompassing conglomeration	Melissa Taylor	Fiction	2010	21.23	44
39	Optimized national process improvement	Megan Goodwin	Fiction	1978	10.99	42
40	Adaptive didactic interface	Natalie Gonzalez	Fiction	1923	25.97	94
47	Reverse-engineered directional conglomeration	John Christian	Fiction	2006	20.37	90
62	Re-contextualized real-time strategy	Nicole Lynch	Fiction	1953	26.34	23
63	Polarized heuristic database	Franklin Mack	Fiction	1989	22.38	56
100	Synchronized client-server service-desk	James Alvarado	Fiction	1906	49.89	29
116	Multi-tiered foreground contingency	Jamie Gates	Fiction	1938	41.82	50
125	Public-key analyzing Graphic Interface	Abigail Madden	Fiction	1990	32.41	16
130	Realigned context-sensitive pricing structure	Jason Rodriguez	Fiction	2004	6.64	90
134	Polarized bandwidth-monitored throughput	Linda Newman	Fiction	1955	35.72	49
142	Multi-tiered responsive parallelism	Amanda Wilson	Fiction	1940	48.96	11
143	Networked multimedia support	Nancy Goodman	Fiction	2012	43.65	50
144	Future-proofed scalable software	Matthew Wolf	Fiction	1945	19.27	90

2) Find books published after the year 1950:

```
SELECT *
FROM books
WHERE Published_Year > ‘1950’;
```

Book_ID	Title	Author	Genre	Published_Year	Price	Stock
2	Persevering reciprocal knowledge user	Mario Moore	Fantasy	1971	35.80	19
4	Customizable 24hour product	Christopher Andrews	Fiction	2020	43.52	8
5	Adaptive 5thgeneration encoding	Juan Miller	Fantasy	1956	10.95	16
6	Advanced encompassing implementation	Bryan Morgan	Biography	1985	6.56	2
8	Persistent local encoding	Troy Cox	Science Fiction	2019	48.99	84
9	Optimized interactive challenge	Colin Buckley	Fantasy	1987	14.33	70
10	Ergonomic national hub	Samantha Ruiz	Mystery	2015	24.63	25
11	Secured zero tolerance time-frame	Denise Barnes	Fantasy	1998	35.95	10
12	Polarized optimal array	Destiny Scott	Non-Fiction	1989	27.43	63
15	User-friendly motivating strategy	Keith Smith	Non-Fiction	1997	23.83	58
17	Reduced secondary core	Benjamin Peters	Fantasy	1966	5.37	45
18	Adaptive 4thgeneration concept	Hector Palmer	Non-Fiction	2021	39.47	32
19	Progressive asymmetric Internet solution	Sean Miller	Science Fiction	1990	11.31	1
20	Face-to-face systematic throughput	Teresa Brennan	Non-Fiction	1978	48.13	64
23	Reverse-engineered context-sensitive ...	Christina Hernandez	Mystery	1967	38.55	70
25	Devolved mobile conglomeration	Alexander Bailey	Biography	1984	8.55	79
26	Multi-channeled multi-tasking capability	Patricia Buck	Science Fiction	1964	21.05	41
30	Multi-layered global open system	Jose Meyer	Biography	2012	30.58	37

3) List all customers from the Canada:

```
SELECT *
FROM Customers
WHERE Country = 'Canada';
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wra

Customer_ID	Name	Email	Phone	City	Country
38	Nicholas Harris	christine93@perkins.com	1234567928	Davistown	Canada
415	James Ramirez	robert54@hall.com	1234568305	Maxwelltown	Canada
468	David Hart	stokesrebecca@gmail.com	1234568358	Thompsonfurt	Canada
NULL	NULL	NULL	NULL	NULL	NULL

4) Show orders placed in November 2023:

```
SELECT *
FROM orders
WHERE Order_Date BETWEEN '2023-11-01' AND "2023-11-30";
```

Order_ID	Customer_ID	Book_ID	Order_Date	Quantity	Total_Amount
4	433	343	2023-11-25	7	301.21
19	496	60	2023-11-17	9	316.26
75	291	375	2023-11-30	5	170.75
132	469	333	2023-11-22	7	194.32
137	474	471	2023-11-25	8	363.04
163	207	384	2023-11-23	3	101.76
182	129	293	2023-11-01	7	125.51
200	313	303	2023-11-23	1	6.57
213	325	447	2023-11-17	7	253.75
231	22	384	2023-11-11	1	33.92
245	386	97	2023-11-01	9	411.66
252	405	387	2023-11-15	5	237.10
257	123	403	2023-11-06	1	15.01
288	6	128	2023-11-13	1	24.04
307	368	133	2023-11-17	1	20.96
322	270	112	2023-11-08	2	16.04

253.75

5) Retrieve the total stock of books available:

```
SELECT SUM(stock) AS TOTAL_STOCK
FROM books;
```

Result Grid | Filter

TOTAL_STOCK
25056

6) Find the details of the most expensive book:

```
SELECT *
FROM books
ORDER BY Price DESC
LIMIT 1;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Co

Book_ID	Title	Author	Genre	Published_Year	Price	Stock
340	Proactive system-worthy orchestration	Robert Scott	Mystery	1907	49.98	88
NULL	NULL	NULL	NULL	NULL	NULL	NULL

7) Show all customers who ordered more than 1 quantity of a book:

```
SELECT *
FROM orders
WHERE Quantity > 1;
```

Order_ID	Customer_ID	Book_ID	Order_Date	Quantity	Total_Amount
1	84	169	2023-05-26	8	188.56
2	137	301	2023-01-23	10	216.60
3	216	261	2024-05-27	6	85.50
4	433	343	2023-11-25	7	301.21
5	14	431	2023-07-26	7	136.36
6	439	119	2024-10-11	5	249.40
7	195	467	2023-10-23	6	82.92
8	32	159	2024-05-07	4	144.84
9	109	407	2024-01-04	9	379.71
10	94	122	2024-07-09	4	123.00
12	454	3	2024-06-17	2	31.50
13	420	180	2023-06-08	5	125.45
14	454	319	2023-08-24	2	85.22
15	127	479	2023-01-10	6	229.62
16	412	196	2023-10-06	8	53.52
17	462	481	2023-03-20	5	52.75

8) Retrieve all orders where the total amount exceeds $20:

```
SELECT *
FROM orders
WHERE Total_Amount > 20;
```

Order_ID	Customer_ID	Book_ID	Order_Date	Quantity	Total_Amount
1	84	169	2023-05-26	8	188.56
2	137	301	2023-01-23	10	216.60
3	216	261	2024-05-27	6	85.50
4	433	343	2023-11-25	7	301.21
5	14	431	2023-07-26	7	136.36
6	439	119	2024-10-11	5	249.40
7	195	467	2023-10-23	6	82.92
8	32	159	2024-05-07	4	144.84
9	109	407	2024-01-04	9	379.71
10	94	122	2024-07-09	4	123.00
11	131	206	2023-10-16	1	38.01
12	454	3	2024-06-17	2	31.50
13	420	180	2023-06-08	5	125.45

9) List all genres available in the Books table:

```
SELECT DISTINCT Genre
FROM books;
```

Result Grid

Genre
Biography
Fantasy
Non-Fiction
Fiction
Romance
Science Fiction
Mystery

10) Find the books with the lowest stock:

```
SELECT *
FROM books
ORDER BY Stock
LIMIT 5;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

Book_ID	Title	Author	Genre	Published_Year	Price	Stock
378	Future-proofed heuristic function	Samantha Mcclain	Romance	1903	6.01	0
44	Networked systemic implementation	Ryan Frank	Science Fiction	1965	13.55	0
60	Robust eco-centric capacity	Brian Haney	Biography	1990	35.14	0
127	Business-focused real-time benchmark	David Nelson	Science Fiction	1997	11.66	0
163	Object-based eco-centric challenge	Douglas Mccarthy	Non-Fiction	1905	19.11	0

11) Calculate the total revenue generated from all orders:

```
SELECT SUM(Total_Amount) AS Revenue_Generated
FROM orders;
```

Revenue_Generated
75628.66

Advance queries:

1) Retrieve the total number of books sold for each genre:

```
SELECT B.GENRE, SUM(O.QUANTITY) AS
TOTAL_NUMBER_OF_BOOKS
FROM BOOKS B
JOIN ORDERS O
ON B.BOOK_ID = O.BOOK_ID
GROUP BY B.GENRE;
```

GENRE	TOTAL_NUMBER_OF_BOOKS
Biography	285
Fantasy	446
Science Fiction	447
Mystery	504
Romance	439
Non-Fiction	351
Fiction	225

2) Find the average price of books in the "Fantasy" genre:

```
SELECT AVG(PRICE) AS AVERAGE_PRICE
FROM BOOKS
WHERE GENRE = 'FANTASY';
```

AVERAGE_PRICE
25.981690

3) List customers who have placed at least 2 orders:

```
SELECT O.CUSTOMER_ID , C.NAME, COUNT(O.ORDER_ID) AS
TOTAL_ORDER
FROM CUSTOMERS C
JOIN ORDERS O
ON C.CUSTOMER_ID = O.CUSTOMER_ID
GROUP BY O.CUSTOMER_ID, C.NAME
HAVING COUNT(ORDER_ID) >= 2;
```

CUSTOMER_ID	NAME	TOTAL_ORDER
2	Crystal Clements	2
6	Stephen Vasquez	2
8	Matthew Johnson	2
13	Kristine Kim	2
14	John [illegible]	2
15	Vane[illegible]	2
16	Stacey Flores	3
21	Edgar Frost	2
22	Stacey Adams	3
23	Hannah Drake	2
24	Christina Mitchell	2
37	Ryan Lara	2
47	Debra Park	3
54	Rachel Jones	2
57	Nicolas Joseph	2
59	Andrew Collins	3

John Wood

4) Find the most frequently ordered book:

```
SELECT O.BOOK_ID, B.TITLE, COUNT(O.ORDER_ID) AS
TOTAL_COUNT
FROM BOOKS B
JOIN ORDERS O
ON B.BOOK_ID = O.BOOK_ID
GROUP BY O.BOOK_ID, B.TITLE
ORDER BY TOTAL_COUNT DESC LIMIT 7;
```

Result Grid | Filter Rows: | Export: | Wrap Ce

BOOK_ID	TITLE	TOTAL_COUNT
31	Implemented encompassing conglomeration	4

5) Show the top 3 most expensive books of 'Fantasy' Genre :

```
SELECT *
FROM BOOKS
WHERE GENRE = 'FANTASY'
ORDER BY PRICE DESC
LIMIT 3;
```

Book_ID	Title	Author	Genre	Published_Year	Price	Stock
240	Stand-alone content-based hub	Lisa Ellis	Fantasy	1957	49.90	41
462	Innovative 3rdgeneration database	Allison Contreras	Fantasy	1988	49.23	62
238	Optimized even-keeled analyzer	Sherri Griffith	Fantasy	1975	48.97	72

6) Retrieve the total quantity of books sold by each author:

```
SELECT B.AUTHOR, SUM(O.QUANTITY) AS TOTAL_QUANTITY
FROM BOOKS B
JOIN ORDERS O
ON B.BOOK_ID = O.BOOK_ID
GROUP BY B.AUTHOR;
```

AUTHOR	TOTAL_QUANTITY
Margaret Moore	8
John Davidson	13
Christopher Fuentes	6
Marissa Smith	16
Christopher Dixon	15
Tonya Saunders	21
Larry Hunt	6
Brandon Foster	4
Michelle Bell	11
Mary French	14
Lisa Lopez	1
Derrick Howard	
David Rodriguez	5
Michael Mckenzie	12
Christian Morales	12

Derrick Howard

7) List the cities where customers who spent over $30 are located:

```
SELECT C.CITY, O.TOTAL_AMOUNT
FROM CUSTOMERS C
JOIN ORDERS O
ON C.CUSTOMER_ID = O.CUSTOMER_ID
WHERE O.TOTAL_AMOUNT > 30;
```

CITY	TOTAL_AMOUNT
Lake Paul	188.56
North Keith	216.60
Kelseyfort	85.50
East David	301.21
Richardsonville	136.36
Ramosstad	249.40
Rogersborough	82.92
New Carlosbury	144.84
Ravenberg	379.71
West Anthony	123.00
North Carolyn	38.01
Micheleborough	31.50
North Joseph	125.45
Micheleborough	85.22
Lake Mary	229.62
Lake Karen	53.52

8) Find the customer who spent the most on orders:

```
SELECT C.CUSTOMER_ID,C.NAME, SUM(O.TOTAL_AMOUNT) AS
TOTAL_SPENT
FROM CUSTOMERS C
JOIN ORDERS O
ON C.CUSTOMER_ID = O.CUSTOMER_ID
GROUP BY C.CUSTOMER_ID, C.NAME
ORDER BY TOTAL_SPENT DESC
LIMIT 1;
```

CUSTOMER_ID	NAME	TOTAL_SPENT
457	Kim Turner	1398.90

457

9) Calculate the stock remaining after fulfilling all orders:

```
SELECT b.book_id, b.title, b.stock, COALESCE(SUM(o.quantity),0) AS
Order_quantity,
b.stock- COALESCE(SUM(o.quantity),0) AS Remaining_Quantity
FROM books b
LEFT JOIN orders o ON b.book_id=o.book_id
GROUP BY b.book_id ORDER BY b.book_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

book_id	title	stock	Order_quantity	Remaining_Quantity
1	Configurable modular throughput	100	3	97
2	Persevering reciprocal knowledge user	19	0	19
3	Streamlined coherent initiative	27	5	22
4	Customizable 24hour product	8	0	8
5	Adaptive 5thgeneration encoding	16	8	8
6	Advanced encompassing implementation	2	0	2
7	Open-architected exuding structure	95	5	90
8	Persistent local encoding	84	3	81
9	Optimized interactive challenge	70	0	70
10	Ergonomic national hub	25	1	24
11	Secured zero tolerance time-frame	10	5	5
12	Polarized optimal array	63	0	63
13	Adaptive 5thgeneration orchestration	99	9	90
14	Re-engineered demand-driven parallelism	95	0	95
15	User-friendly motivating strategy	58	0	58
16	Vision-oriented tangible project	8	1	7

Result 13